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RBE211TC DYNAMIC SYSTEMS

MODELLING ELECTRICAL AND ELECTROMECHANICAL SYSTEMS

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Lecture Outline

1. Review of Electrical Circuit Elements
2. Kirchhoff's Laws
3. Modelling Electrical Systems
4. Electromechanical Systems

Motivation

- Engineering systems often combine multiple physical domains
- Examples:
  - Electric motors in robotic joints
  - Electromagnetic actuators and solenoid valves
  - Motor-driven mechanisms in mobile robots
  - Sensors and control circuits interacting with mechanical systems
- Goal:

  Develop mathematical models that describe the dynamics of electrical and electromechanical systems.

Modelling Across Physical Domains

- Mechanical Systems:
  - Mass
  - Spring
  - Damper
- Electrical Systems:
  - Inductor
  - Capacitor
  - Resistor
- Many engineering systems combine electrical and mechanical components.
- In robotics, actuators and sensors often combine electrical and mechanical elements.

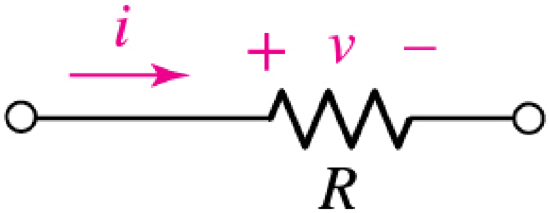
Resistor

- A resistor relates voltage and current by Ohm's law

$$v = iR$$

where R is the resistance.

- It is an energy-dissipating element.
- The unit of resistance is ohm (Ω).



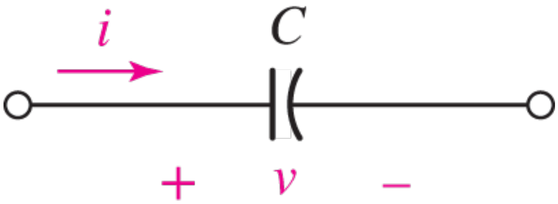
Capacitor

- A capacitor relates current and voltage by

$$i = C\frac{dv}{dt}$$

where C is the capacitance.

- It is an energy-storing element.
- The unit of capacitance is farad (F).
- A capacitor stores energy in an electric field.

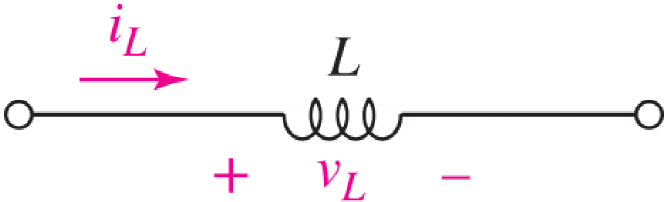


Inductor

- An inductor relates voltage and current by

$$v = L\frac{di}{dt}$$

  where L is the inductance.
- It is an energy-storing element.
- The unit of inductance is henry (H).
- An inductor stores energy in a magnetic field.



Summary of Basic Electrical Elements

Element	Constitutive Law	Role
Resistor	$v = iR$	Dissipates energy
Capacitor	$i = C\frac{dv}{dt}$	Stores energy
Inductor	$v = L\frac{di}{dt}$	Stores energy

- Resistors are dissipative elements, whereas capacitors and inductors are energy-storage elements.

Power and Energy in Electrical Elements

- Resistor: dissipates electrical energy as heat

$$P_R = vi = i^2 R = \frac{v^2}{R}$$

- Capacitor: stores energy in an electric field

$$E_C = \frac{1}{2} C v^2$$

- Inductor: stores energy in a magnetic field

$$E_L = \frac{1}{2} L i^2$$

- Dynamic variables are associated with energy-storage elements.

Kirchhoff's Laws

Two fundamental laws are used to model electrical circuits:

- Kirchhoff's Voltage Law (KVL)

  The algebraic sum of voltages around a closed loop is zero.

$$\sum v = 0$$

- Kirchhoff's Current Law (KCL)

  The algebraic sum of currents at a node is zero.

$$\sum i = 0$$

Two-Step Modelling Process

Step 1 – Physical Modelling

- **Identify the circuit elements**
- Define voltages and currents
- Select the dynamic variable(s)

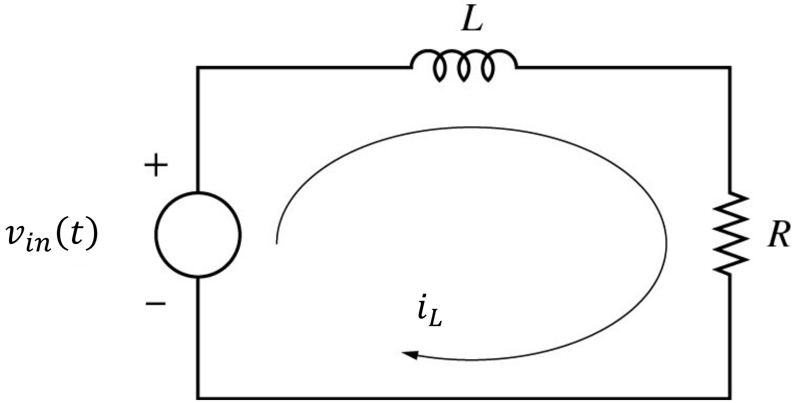
Step 2 – Mathematical Modelling

- Apply KVL, KCL, mesh analysis, or nodal analysis
- Substitute the element laws
- Obtain the governing differential equation(s)

Dynamic variables are associated with the energy-storage elements, such as capacitor voltage and inductor current.

Example 1

Figure below shows a series RL circuit driven by a voltage source $v_{in}(t)$. Derive the mathematical model of the electrical system.



Example 1

Example 1

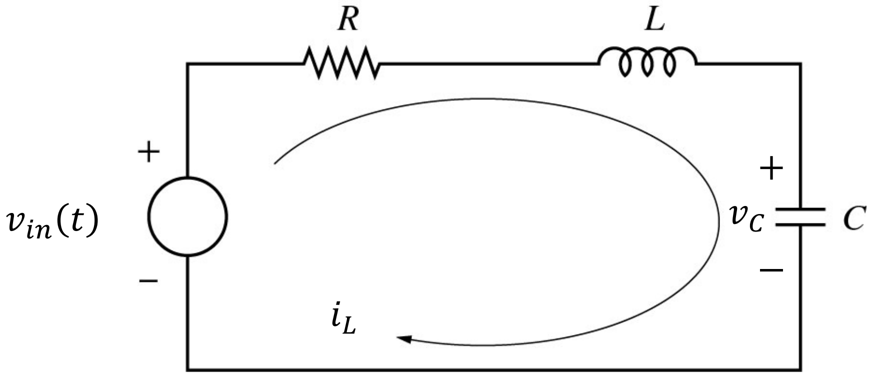
This is a linear first-order ODE.

- Dynamic variable: $i_L(t)$
- Input: $v_{in}(t)$

Because the circuit contains only one energy-storage element, the system is first order.

Example 2

Figure below shows a series RLC circuit driven by a voltage source $v_{in}(t)$. Derive the mathematical model of the electrical system.



Example 2

Example 2

Together with

$$C\frac{dv_C(t)}{dt} = i_L(t)$$

the mathematical model can be written as:

This is a set of two coupled first-order ODEs.

Example 2

Using

$$i_L(t) = C\dot{v}_C(t)$$

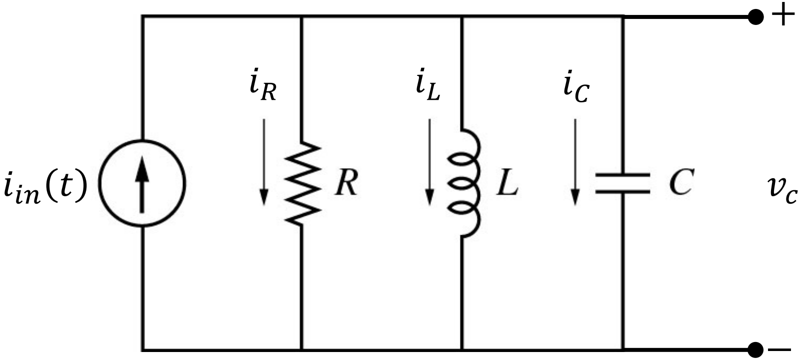
and substituting into:

This is a linear second-order ODE with dynamic variable $v_C(t)$ and input $v_{in}(t)$.

This has the same mathematical structure as the mechanical mass-spring-damper model discussed previously.

Example 3

Figure below shows a parallel RLC circuit driven by a current source $i_{in}(t)$. Derive the mathematical model of the electrical system.



Example 3

Apply Kirchhoff's Current Law at the top node:

$$i_R(t) + i_L(t) + i_C(t) = i_{in}(t)$$

Since all parallel elements share the same voltage, substitute the element laws:

$$i_R(t) = \frac{v_C(t)}{R}, \qquad i_C(t) = C\frac{dv_C(t)}{dt}$$

and for the inductor:

$$v_C(t) = L\frac{di_L(t)}{dt}$$

Hence,

$$\frac{v_C(t)}{R} + i_L(t) + C\frac{dv_C(t)}{dt} = i_{in}(t)$$

Example 3

Rearrange gives

$$C\frac{dv_C(t)}{dt} + \frac{1}{R}v_C(t) + i_L(t) = i_{in}(t)$$

Together with

$$L\frac{di_L(t)}{dt} = v_C(t)$$

the mathematical model can be written as:

$$\boxed{\begin{gathered} C\dot{v}_C(t) + \frac{1}{R}v_C(t) + i_L(t) = i_{in}(t) \\ L\,\dot{i}_L(t) - v_C(t) = 0 \end{gathered}}$$

This is a set of two coupled first-order ODEs.

Example 3

Using

$$L\,\dot{i}_L(t) = v_C(t)$$

This is a linear second-order ODE with dynamic variable $v_C(t)$ and input $\dot{i}_{in}(t)$.

Electromechanical Systems

- Electromechanical systems combine electrical and mechanical subsystems.
- They convert:
  - electrical energy into mechanical motion
  - mechanical motion into electrical signals
- Common examples include:
  - DC motors
  - solenoids
  - generators
  - electromagnetic actuators
- In robotics, electromechanical systems are widely used in joints, wheels, grippers, and drive systems.

Current-Magnetic Field Interaction

Three key ideas:

1. An electrical current establishes a magnetic field
2. A current-carrying conductor in a magnetic field experiences a force
3. A moving conductor in a magnetic field has an induced voltage

These relationships form the physical basis of many electromechanical systems.

Electromechanical Coupling in a DC Motor

Two key coupling relationships:

Motor torque

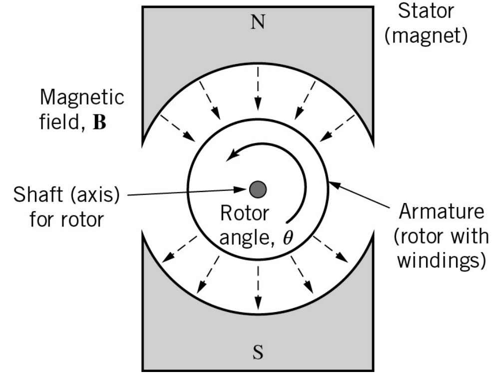
$$\tau_m = K_m i_a$$

Induced voltage (back-emf)

$$v_b = K_b \dot{\theta}$$

Where:

- i_a = armature current
- $\dot{\theta}$ = angular velocity of the rotor
- K_m = motor torque constant
- K_b = back-emf constant



Stator and armature of a DC motor.

Modelling Electromechanical Systems

Step 1 – Electrical subsystem

- Apply KVL and the electrical element laws
- Obtain the armature-circuit equation

Step 2 – Mechanical subsystem

- Apply Newton's second law for rotation
- Obtain the rotor-dynamics equation

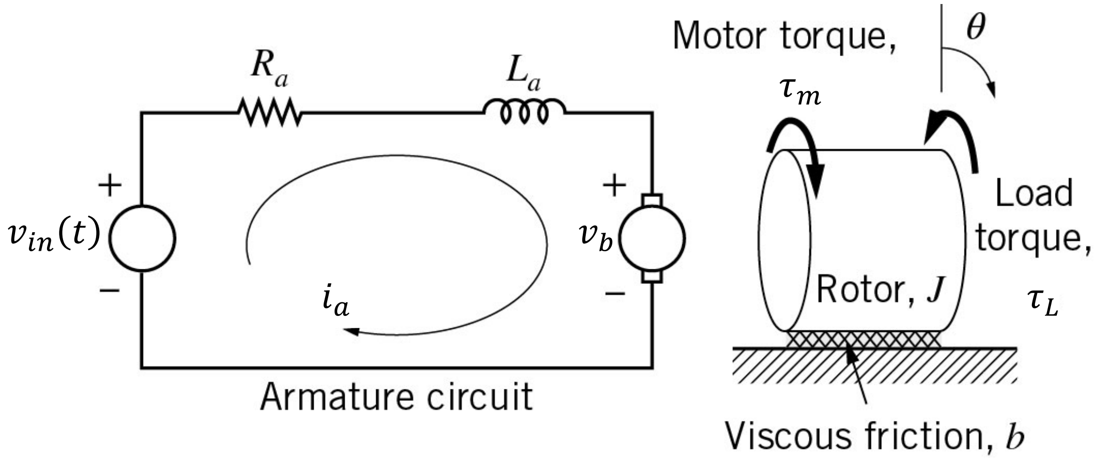
Coupling

- Use $\tau_m = K_m i_a$ and $v_b = K_b \dot{\theta}$ to connect the two subsystems.

Result: a set of coupled differential equations for the complete system

Example 4

Figure below shows a simplified DC motor model. Derive the mathematical model of the electromechanical system.



Example 4

Apply KVL to the armature circuit:

Example 4

Apply Newton's Second Law for rotation:

Example 4

Therefore, the mathematical model of the DC motor is

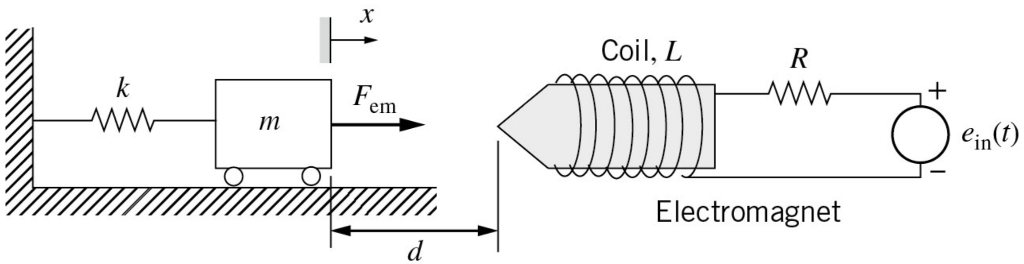
This is a set of coupled differential equations.

- Electrical dynamic variable: $i_a(t)$
- Mechanical response variable: $\theta(t)$
- Inputs: $v_{in}(t)$, $\tau_L(t)$

The armature current generates motor torque, while rotor speed generates back emf.

Solenoid Actuator

- A solenoid actuator converts electrical energy into linear motion.
- It typically consists of a coil, a movable plunger / armature, a return spring, and a mechanical load.



- The electrical input produces current in the coil, which generates an electromagnetic force. That force drives the plunger linearly.
- Common applications: valves, relays, locks, switching mechanisms, robotic gripping and release devices

Solenoid Models Are Often Nonlinear

In a solenoid actuator:

- the inductance may depend on plunger position:

$$L = L(x)$$

- the electromagnetic force depends on current and position
- therefore, the electrical and mechanical equations are often nonlinear

Typical modelling structure:

- Electrical subsystem: coil
- Mechanical subsystem: mass-spring-damper
- Coupling: electromagnetic force, F_{em}

In practice:

- Nonlinear models are often linearized about an operating point
- The linearized model is useful for analysis and control design

Review

1. Electrical Circuit Elements
2. Kirchhoff's Laws
3. Mathematical Modelling of Electrical Systems
4. Electromechanical Systems
5. DC Motor Modelling
6. Solenoid Actuator and Nonlinear Modelling